

HEALIX

AI-Powered Personalized Health & Lifestyle Assistant

1. PROJECT VISION

Problem

People with chronic lifestyle conditions like:

- Polycystic Ovary Syndrome
- Type 2 Diabetes
- Hypothyroidism
- obesity
- hypertension

need personalized lifestyle management.

Currently they must use:

- one app for calorie tracking
- another for workouts
- ChatGPT for meal plans
- Google for recipes
- Notes app for progress

Everything is fragmented.

Solution

Healix combines:

- Health profile management
- AI-generated meal plans
- AI-generated workout plans
- Progress tracking
- Condition-specific recommendations
- Habit building

all inside a single platform.

2. MVP SCOPE

For the first version, ONLY support:

Disease : PCOS

because:

- you understand the problem
- huge audience
- enough complexity
- easier dataset collection
- Do NOT add diabetes initially.

3. USER FLOW

Step 1

User signs up

Fields:

- Name
- Email
- Password

Step 2

Profile Setup

Fields:

- Age
- Gender
- Height
- Weight
- Goal Weight
- Activity Level
- Food Preference
- Budget
- Lifestyle

Step 3

Condition Selection

Initially: PCOS

Later:

- Diabetes
- Thyroid
- Hypertension
- Step 4

AI Generates

- Meal Plan
- Workout Plan
- Water Goal
- Sleep Goal

Step 5

Dashboard

User sees:

Today's:

- Meals
- Workout
- Water intake
- Calories

4. COMPLETE SYSTEM ARCHITECTURE

React Frontend

↓

FastAPI Backend

↓

MongoDB Database

↓

Gemini/OpenAI API

FRONTEND MODULES

Authentication

Pages:

- Login
- Signup
- Forgot Password
- Profile Module

Store:

```
{  
  
  "name": "",  
  
  "age": 21,  
  
  "weight": 90,  
  
  "height": 157,  
  
  "condition": "PCOS"  
}
```

Dashboard Module

Cards:

- Today's calories
- Water intake
- Workout
- Weight trend
- Meal Plan Module

Display:

- Breakfast
- Lunch
- Dinner
- Snack
- Calories

Workout Module

Display:

- Exercises
- Sets
- Reps
- Duration

Progress Module

Charts:

- Weight
- Streaks
- Water intake

BACKEND MODULES

Let's organize it properly.

Module 1

Authentication Service

Endpoints:

- POST /signup
- POST /login
- POST /logout

Uses:

JWT Authentication

Module 2

User Service

Endpoints:

- GET /user
- PUT /user
- DELETE /user

Purpose:

Manage profile.

Module 3

Health Profile Service

Endpoints:

- POST /health-profile
- GET /health-profile

Stores:

- weight
- height
- activity level
- condition

Module 4

AI Recommendation Service

Endpoints:

- POST /generate-meal-plan
- POST /generate-workout
- This talks to Gemini/OpenAI.

Flow:

User Data



Prompt Builder



AI API



Response



Database

Module 5

Progress Tracker

Endpoints:

- POST /weight-entry
- GET /weight-history

Stores:

Daily weight logs.

DATABASE DESIGN

Collections:

Users

```
{  
  
  "_id": "",  
  
  "name": "",  
  
  "email": "",  
  
  "password": ""  
}
```

Profiles

```
{  
  
  "_id": "",  
  
  "userId": "",  
  
  "age": 20,  
  
  "height": 157,  
  
  "weight": 100  
}
```

Conditions

```
{  
  
  "_id": "",  
  
  "userId": "",  
  
  "condition": "PCOS"  
}
```

MealPlans

```
{  
  
  "_id": "",
```

```
"userId":"","  
"plan":"..."  
}
```

Workouts

```
{  
  "_id":"",  
  "userId":"",  
  "plan":"..."  
}
```

ProgressLogs

```
{  
  "_id":"",  
  "userId":"",  
  "weight":99,  
  "date":"..."  
}
```

PHASE BREAKDOWN

PHASE 1

Foundation (Week 1)

Learn:

- HTML
- CSS
- JavaScript
- Git
- GitHub

Build:

- Static pages.

- Output:
- Frontend UI.

PHASE 2

React Frontend (Week 2)

Learn:

- React
- Components
- Routing
- State

Build:

- Login
- Signup
- Dashboard UI
-

Output:

Frontend prototype

PHASE 3

FastAPI Backend (Week 3)

Learn:

- FastAPI
- APIs
- JSON
- CRUD

Build:

- Backend server

Output:

Working API

PHASE 4

Database (Week 4)

Learn:

- MongoDB
- Collections
- CRUD

Build:

- Store users

Output:

Persistent data

PHASE 5

Authentication (Week 5)

Learn:

- JWT
- Password Hashing

Build:

- Login system
- Output:
- Protected routes

PHASE 6

AI Integration (Week 6)

Learn:

- Gemini API
- Prompt Engineering

Build:

- Meal generation
- Workout generation

Output:

Personalized recommendations

PHASE 7

Progress Tracking (Week 7)

Build:

- Weight logs
- Progress dashboard

Output:

User analytics.

PHASE 8

Deployment (Week 8)

Deploy:

Frontend:

Vercel

Backend:

Render

Database:

MongoDB Atlas

Output:

Live application.

Future Versions (Version 2)

After MVP works:

Add

- Multiple diseases
- Menstrual cycle tracking
- Grocery list generation
- AI chatbot
- Habit tracker
- Sleep tracking
- Notification system
- Doctor dashboard
- Community section
- Voice assistant